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FACE RECOGNITION BY GEOMETRIC FEATURES

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Abstract

We presented a part of our study of the face recognition problem. A feature points extraction by template matching and Gabor filtering has been described. A geometric approach to face recognition based on the feature points was tested. The results of human photo recognition using a feature set based on the points are presented.

Keywords: Face recognition, feature point extraction, geometric features, template matching, Gabor filtering

INTRODUCTION

There are a lot of approaches to face recognition problem [1,2,4,6]. One of them is based on the face feature points. Our input is digital images of frontal face portrait. We need 30 - 40 points to describe a face robustly. Locations of some points depend on face expression. Such points are not very good for recognition. There are two problems: to define and extract the most invariant points and to find the optimal geometric feature set for face recognition. Images of similar people or images of one person must be closer to each other in the feature space than to images of other persons. The paper presents our work to solve the problems mentioned above.

1. FACIAL KEYPOINT LOCATIONS

1.1. Template matching

To find facial keypoints, we have tested a technique based on the use of template prototypes, (which are portrayed as grayscale images). Correlation as a basic measure for comparing templates with a part of the image that we wish to recognize. Image normalization is required in order for common reference to exist, since template matching is significantly affected by variances in scaling, illumination and rotation. In [2] faces are portrayed in a database as images of the whole head and as four supplementary different masks that isolate regions of eyes, nose, mouth and face (region that is defined by the eyebrows, the chin and the left and the right sides of a head). These regions are of the same scale for all face images. In such cases, recognition involves comparison of a template face image in question with all the face images included in the database. The output of the locating system is a vector that contains the correlation coefficients for all the regions. A face in the database is recognized if it has the maximal sum of the partial correlation coefficients with the template image.

Figure 1. Several templates used in the presented approach.

Another question is to study behavior of the recognition procedure with templates in different resolutions. Gaussian pyramids were used [2]. This involves multiresolution template representations. Our experiments proved that not very high resolutions are needed for template matching.

One may range characteristics of a face in a way that depicts how well each of them assists to recognition. Eyes proved to be the most significant feature. Next nose comes, then mouth and face as a whole. This fact is explainable, since the face as a whole is vulnerable to variations of the frontal views and due to imperfect normalization.

1.2. Gabor filtering

Gabor filters may be used in a facial recognition system also. The neighboring region of a pixel may be described by the response of a group of Gabor filters in different frequencies and directions, which have a reference to a specific pixel. In that way, a feature vector may be formed, containing the responses of those filters. 2-d form of the Gabor filter is the following:

$$g(x,y,u,v) = \exp\left(-\left(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}\right) + 2\pi j\left(ux + vy\right)\right),$$

Where x, y is a point location, u, v govern the filter's central spatial frequency, which determines its orientation, and the parameters s control the width of the Gaussian window relative to the wavelength corresponding to the central frequency [3]. The feature vector consists of the responses of the filters in different central frequencies and for a specific position. In comparison of two images, the phase difference of the response of each filter corresponds to local variation towards the filter's direction. These local position variations are combined to express the 'distance' between two images.

A wavelet representation of a Gabor filter is the following:

$$\Phi_\lambda(x,y,\theta) = \exp\left\{-\lambda^2 \cdot (x^2 + y^2) + j\pi \cdot (x \cdot \cos\theta + y \cdot \sin\theta)\right\}$$

The parameter θ shows the filter's orientation in space. Orientation has to do with feature extraction towards a desired direction. The filter's orientation is strongly connected to its central frequency.

2. FACIAL KEYPOINT DETECTION

2.1. Automated facial feature extraction

In our approach, the automated detection of the predetermined points consists of two steps: the hierarchic and reliable selection of specific image blocks and subsequently application of a standardized procedure for the points.

In order for the former of the two processes to be successful, the need of a secure method of approach emerged. Detection of a block describing a facial feature relies on a previously effectively detected feature. By adopting this reasoning, the choice of the most significant characteristic – the ground of the cascade routine – had to be made. Importance of the commonly used facial features, regarding the issue of face recognition, has already been studied [1]. The outcome of surveys proved eyes to be the most dependable and easily located of all facial features, and as such they were used. The techniques that were developed and tried separately utilize a combination of template matching and Gabor filtering [5].

After having isolated the restricted regions of interest from a frontal image, localization of the predetermined points ensues. The center of the eye's pupil is searched as the darkest point both at the integrated horizontal and vertical direction of the eye's block. The exact position of nostrils is sought from the sides to the center of the nose block. Mouth tips are met in a similar manner. The right and left head edges at the altitude of eyes are retrieved from the horizontally integrated vector describing the area where the template hair starts. Similar heuristics we apply for other point detection. The whole search procedure is attempted to be as close to human perception as possible.

2.2. The hybrid method

The basic quest of the desired feature blocks is performed by a simple template matching procedure. Each feature prototype is selected from one of the frontal images of the face database. The practiced comparison criterion is the maximum correlation coefficient between the prototype and the repeatedly audited blocks of a smartly restricted area of the face.

In order for the search area to be incisively and functionally limited, the knowledge of the human face physiology has been applied, without hindering the satisfactory performance of the algorithm in cases of small violations of the initial limitations. The final block selection by the mere use of this method has not always been crowned with success. For that reason, usage of Gabor filtering was deemed to be one suitable tool. As it can be mathematically deduced from the filter's form, it ensures simultaneous optimum localization in the spatial space as well as in frequency space. The filter is applied both on the localized area and the template in four different spatial frequencies. Its response is regarded as valid, only in the case that its amplitude exceeds a

saliency threshold. The area with minimum phase distance from its template is considered to be the most reliably traced block.

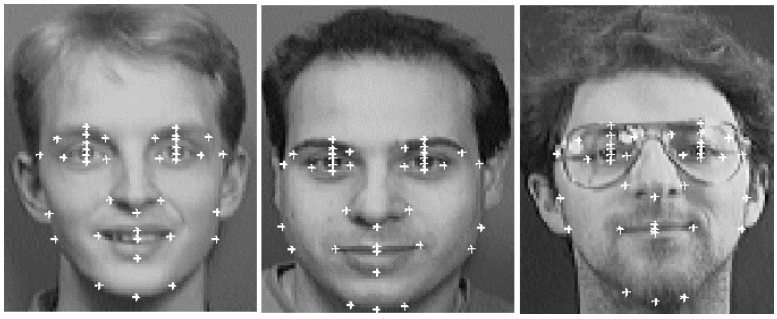

Figure 3. Points detected in experiments.

2.3. Geometric features

We extracted 37 points in experiments. There are a lot of geometric features based on the points. Geometric features may be presented by segments, perimeters and areas of some figures formed by the points. How to choose the optimal feature set? To compare the recognition results we studied the feature set described in [6]. It includes 15 segments between the points and the mean values of 15 symmetrical segment pairs. We tested different subsets of the features to looking for the most important features. Details are described in [7]. For every feature set we calculated:

$$F = \sqrt{\frac{\sum_{i=1}^{k}\left(M_{D_i} - D_i\right)^2}{\sum_{i=1}^{k}\left(M_{M_i} - M_i\right)^2}} \quad ,$$

where M_i and D_i are mean and variance of the feature values for k images of the i-th person, M_{Di} and M_{Mi} are mean of D_i and M_i, respectively. The lowest F value corresponds to the better feature set.

We have selected 28 features and simultaneously improved the recognition rate up to 98,5%.

3. EXPERIMENTS

The Olivetti Research Laboratory face database was used in experiments. An example of our recognition is presented in Fig. 4. It may be seen that in spite of small rotation, orientation and illumination variances, the algorithm works in a fairly robust manner. Each photo is an image of the size of 92×112 pixels, and quantized to 256 gray levels. We tested 70 images of 12 persons. Distances in the feature space from a template image to every image in the database were calculated. The 5 nearest images were derived and if there were photos of the query person then the result was considered positive (Fig. 4). Each image was tested as a query and compared with

others. Just in one case of 70 tests there were no any image of the query person through the 5 nearest ones, i.e. the recognition rate was 98.5%. Note, the best rate in study [6] applied to the same database was 89,36 % on about 100 images.

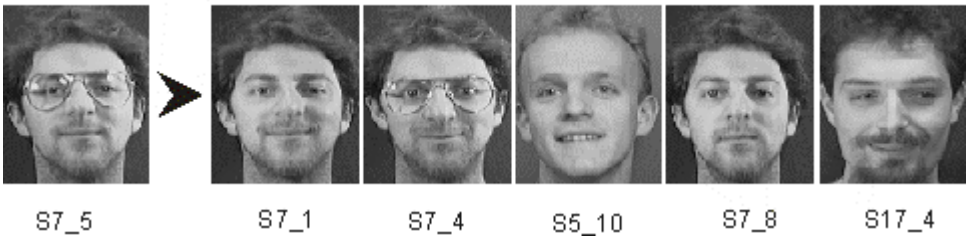

Figure 4. Five images which are nearest to a template in accordance to our approach.

4. CONCLUSION

The paper presents results of the face recognition study based on geometric features. A way of anthropometrical point extraction is described. Simple features based on the points give good recognition rate. Experimental results demonstrate the geometric features based approach can be effectively explored for preliminary (coarse) face recognition. Then for final recognition more precise techniques may be applied.

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